

# Trajectory-Aligned Latent Handoff for Efficient High-Resolution Video Diffusion

Anonymous submission

## Abstract

High-resolution video diffusion is costly because every denoising step processes long spatiotemporal latent sequences at the target resolution. Since early steps mainly determine global structure and motion while later steps recover details, mixed-resolution sampling can reduce computation; however, resolution handoff must address residual trajectory error, non-scale-equivariant VAE latents, and high-resolution trajectory re-entry. We propose **Trajectory-Aligned Latent Handoff (TALH)**, a lightweight latent-space framework that performs most denoising at low resolution and decomposes the handoff into three components: a LoRA-based **Trajectory Alignment Adapter (TAA)** that calibrates the residual trajectory gap; a **Clean Latent Lifter (CLL)** that maps clean latents across resolutions; and **High-Resolution Trajectory Re-entry (HTR)** to re-noise the lifted latent for continued high-resolution denoising. Operating under a model-endogenous self-distillation paradigm, TAA and CLL are trained from supervision generated by the frozen target model, requiring no external paired videos or additional teacher. On 50-step Wan2.1 with a  $368 \times 640 \rightarrow 720 \times 1248$  handoff, TALH-Q and TALH-E achieve  $1.83\times$  and  $2.22\times$  end-to-end speedups while retaining 97.76% and 97.53% of Native-HR VBench-5. Compared to trilinear interpolation, CLL reduces latent  $L_1$  error by 48.6% and LPIPS by 73.3% for cross-resolution lifting, while TAA reduces endpoint  $L_1$  error by 25.8% and 21.0% at steps 40 and 45. TALH consistently exhibits a robust quality-efficiency Pareto frontier across both standard 50-step and 4-step distilled samplers.

## 1 Introduction

Recent latent video diffusion models such as Wan generate semantically coherent videos with realistic motion and fine details (Wan et al. 2025), but their inference cost grows aggressively with spatial resolution, video length, and sampling steps. In DiT-based models, every denoising iteration processes the full spatiotemporal token sequence. Executing the entire trajectory at the target resolution forces early steps—which mainly establish coarse structure and macro-motion—to bear the full burden of high-resolution computation, severely limiting real-time deployment. Since diffusion sampling inherently distributes generation across stages (global layout in early steps, high-frequency textures in later steps), a promising paradigm is mixed-resolution inference: modeling global content in a low-resolution latent space first,

This is an anonymized submission for review purposes only. Distribution, citation, or public sharing of this manuscript is strictly prohibited. Copyright and publication details will appear in the final version if accepted.

then switching to the target resolution for a short refinement suffix. This spatial-scaling approach (ModelTC 2026; Zheng et al. 2026) is fundamentally orthogonal and complementary to temporal step-distillation methods.

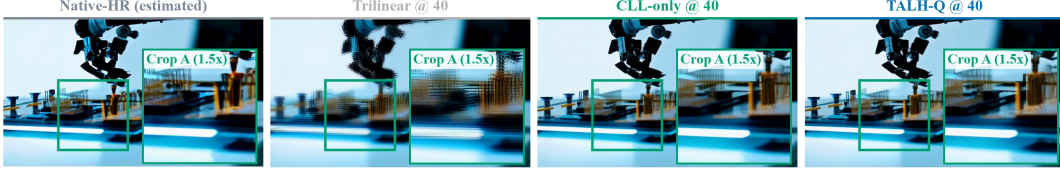
Yet, dynamic resolution handoff is far more complex than simple tensor resizing, collectively dictated by three error sources: trajectory gap, scale discrepancy, and refinement residual. First, a substantial residual denoising gap persists between the single-step prediction and the completed low-resolution endpoint. Second, video VAE latents lack strict scale equivariance, failing to capture the intricate cross-resolution correspondences jointly determined by the VAE architecture and scale transitions. Finally, the lifted latents must seamlessly re-enter the high-resolution trajectory for texture refinement. Directly mapping the handoff prediction to the high-resolution endpoint forces a single Joint Trajectory-Scale Lifter (JTSL) to resolve these challenges jointly. Because the trajectory gap fluctuates with timesteps, schedulers, and prompts, and since high-resolution details are underdetermined by low-resolution inputs, such joint regression heavily suffers from high-frequency attenuation. Our handoff-step scans confirm this, motivating a decomposition into localized subproblems.

We therefore propose **TALH** (Trajectory-Aligned Latent Handoff), comprising three modular components activated at the handoff step: a LoRA-based **Trajectory Alignment Adapter (TAA)** to calibrate the residual trajectory gap; a **Clean Latent Lifter (CLL)** to map cross-scale clean latents; and a **High-Resolution Trajectory Re-entry (HTR)** protocol for generative refinement. Both TAA and CLL are trained via a model-endogenous self-distillation framework supervised directly by the frozen target generator.

Our primary contributions are summarized as follows:

- **We propose TALH**, a lightweight mixed-resolution framework that shifts the bulk of denoising computations to a low-resolution prefix, providing controllable quality-efficiency Pareto operating points (TALH-Q/E) for online inference.
- **We decouple dynamic resolution handoff** into three explicit phases via TAA, CLL, and HTR. This mathematical decomposition significantly mitigates the epistemic uncertainty inherent in joint trajectory-scale mapping, enabling isolated diagnostics.

(a) Matched real video frames — prompt 05, seed 9705, frame 41



(b)

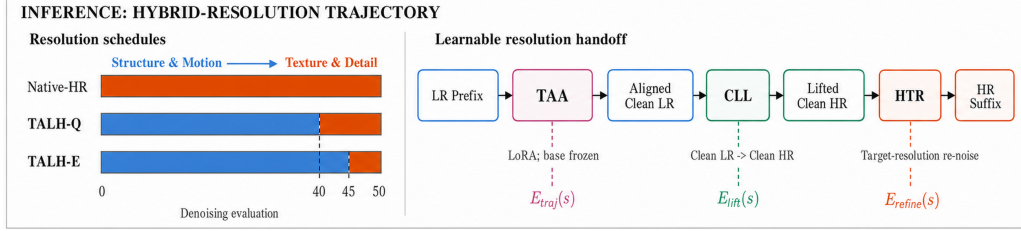


Figure 1: Real-frame comparison and TALH inference. (a) Content-aligned Native-HR (estimated), Trilinear, CLL-only, and TALH-Q outputs for prompt 05, seed 9705, frame 41, with identical  $1.5\times$  crops. The reference follows a complete 368p trajectory; handoff outputs are 720p. (b) Native-HR performs all 50 evaluations at high resolution, whereas TALH-Q and TALH-E execute the first 40 and 45 evaluations at low resolution and retain 10 and 5 high-resolution suffix evaluations.

- We construct a model-endogenous training paradigm requiring no external data or teachers, and establish the consistency condition for cached-prefix states. Experiments on standard and distilled Wan samplers validate its robust acceleration and schedule compatibility.

## 2 Related Work

### 2.1 Latent Video Generation and Cascaded Upscaling

Latent Diffusion Models compress the generative process into a compact space for scalable high-resolution video synthesis (Rombach et al. 2022). Traditional frameworks or recent systems like LTX-2 and LUVe typically employ cascaded pipelines, using independent spatial latent upsamplers or post-generation refinement modules to elevate resolution (Blattmann et al. 2023; HaCohen et al. 2026; Zhao et al. 2026). Similarly, video super-resolution methods focus on restoring already decoded RGB outputs (Zhou et al. 2024; Xu et al. 2024).

Unlike these approaches that treat upscaling as an isolated post-processing stage, TALH natively embeds the Clean Latent Lifter (CLL) into the original Wan sampling trajectory. By linking the lifted latents to subsequent steps via High-Resolution Trajectory Re-entry (HTR), cross-resolution scaling and generative priors are unified within a single, continuous inference process.

### 2.2 Mixed-Resolution Diffusion Inference

The changing-resolution inference in LightX2V utilizes fixed interpolation to adjust latent sizes mid-sampling (ModelTC 2026). Concurrently, RALU demonstrates that training-free latent upscaling suffers from high-frequency aliasing and a resolution-dependent noise-timestep mismatch, designing a mixed-resolution scheme for Diffusion Transformers (Jeong et al. 2025). For image flow matching, MrFlow combines low-resolution structure generation with low-noise high-resolution refinement (Zheng et al. 2026). TALH extends this

paradigm to video trajectories by learning cross-resolution lifting directly within the model’s latent space, while explicitly rectifying low-resolution tail-trajectory errors prior to the handoff.

### 2.3 Few-Step Distillation and On-Policy Trajectory Learning

To accelerate inference, temporal distillation methods minimize sampling steps via trajectory regression or consistency mapping (Luo et al. 2023; Yin et al. 2024; Zhai et al. 2024). Recent works like Self-Forcing and AnyFlow further emphasize enforcing strict consistency between student training states and actual inference trajectories to mitigate distribution shift (Huang et al. 2025; Gu et al. 2026).

TALH focuses on spatial rather than temporal computation, making it orthogonal to step reduction. Since TAA remains inactive prior to the handoff step, its parameter updates do not alter the low-resolution prefix trajectory. Leveraging this local activation property, Section 3.5 establishes the consistency conditions that allow cached prefix states to be directly accessed during inference without training-inference distribution drift.

## 3 Method

### 3.1 Problem Definition and Overall Framework

Let  $v_\phi(x_s, s, c)$  denote a frozen Wan flow predictor, where  $x_s$  represents the intermediate latent before the  $s$ -th denoising calculation,  $c$  is the text condition, and  $T$  is the total number of denoising steps. We use the superscripts  $L$  and  $H$  to signify the low-resolution and high-resolution latent spaces, respectively. Let  $z_T^L$  denote the endpoint of the complete low-resolution trajectory generated by the frozen model under a given text prompt, random seed, initial noise, and low-resolution scheduler. We define  $(z_0^L, z_0^H)$  as the cross-resolution clean latent pair obtained by encoding the low- and high-resolution versions of the same video through the Wan VAE.

TALH introduces two independently trained modules: the Trajectory Alignment Adapter (TAA) parameterized by LoRA weights  $\Delta\theta_s$ , and the Clean Latent Lifter (CLL) denoted as  $U_\psi^{\text{CLL}}$  with parameters  $\psi$ . High-Resolution Trajectory Re-entry (HTR) is a parameter-free re-routing operation. As illustrated in the upper panel of Figure 2, the complete data flow is formulated as:

$$x_s^L \xrightarrow{\text{TAA: } v_\phi + \Delta\theta_s} \hat{z}_s^L \xrightarrow{U_\psi^{\text{CLL}}} \hat{z}_s^H \xrightarrow{\text{HTR}} x_{s+1}^H.$$

Crucially, TAA is dedicated solely to aligning the current low-resolution clean prediction with the complete trajectory’s low-resolution endpoint, while CLL focuses exclusively on the cross-resolution mapping of clean latents. These two modules are trained separately without joint backpropagation. This decoupled training strategy prevents the massive memory overhead of hosting both the 14B denoiser and the CLL module simultaneously in the compute graph, while enabling an isolated experimental analysis of trajectory and scale errors.

The handoff step simultaneously governs three distinct types of errors. Let  $E_{\text{traj}}(s)$  denote the residual trajectory gap before the handoff,  $E_{\text{lift}}(s)$  be the cross-resolution lifting error, and  $E_{\text{refine}}(s)$  represent the remaining high-resolution error left uncorrected after HTR. The total handoff error can be conceptualized as:

$$E_{\text{handoff}}(s) \approx E_{\text{traj}}(s) + E_{\text{lift}}(s) + E_{\text{refine}}(s).$$

This conceptual decomposition reveals the central trade-off across choices of  $s$ . A later handoff reduces the residual trajectory gap  $E_{\text{traj}}(s)$  and moves the CLL input closer to its clean-latent training distribution, thereby reducing  $E_{\text{lift}}(s)$ . However, a later handoff also leaves fewer high-resolution suffix evaluations and can therefore increase the uncorrected refinement error  $E_{\text{refine}}(s)$ . TAA, CLL, and HTR are designed to address these three error sources separately.

### 3.2 Model-Endogenous Dual-Supervision Data

TALH operates entirely without external paired videos, pre-trained super-resolution weights, or additional teacher models. The supervision for both CLL and TAA is generated natively by the frozen target generator, tracking two distinct conditional distributions as shown in the lower panel of Figure 2.

**Cross-Resolution Lifting Pairs.** We run the frozen Wan model using a set of predefined text prompts and fixed random seeds to generate high-resolution reference videos. These videos are spatially downsampled in the RGB pixel space. Both the high- and low-resolution variants are then encoded using the same Wan VAE to produce perfectly aligned cross-resolution clean latent pairs  $(z_0^L, z_0^H)$ . This construction preserves the frame count, semantic content, and motion trajectories, forcing CLL to learn cross-scale correspondences tightly bound to the target model’s specific VAE and generative distribution.

**Trajectory Alignment Pairs.** Under identical text prompts, random seeds, initial noise, and schedulers, we roll out the complete low-resolution trajectory of the frozen

model. We cache the intermediate low-resolution state  $x_s^L$  immediately preceding the handoff step and utilize the final low-resolution trajectory endpoint  $z_T^L$  as the ground-truth target. Consequently, CLL receives cross-resolution structural supervision, while TAA receives cross-timestep trajectory supervision, forming a unified model-endogenous resolution-trajectory self-distillation framework.

Although both types of supervision share the same base generative model, they map to different conditional distributions: the low-resolution input for CLL is derived from downsampled high-resolution video encodings, whereas TAA directly targets the endpoints of native low-resolution trajectories. Section 4.2 evaluates their end-to-end synergy via a  $2 \times 2$  factorial design crossing trajectory states and resolution operators.

### 3.3 Cross-Resolution Clean Latent Lifting

The Clean Latent Lifter (CLL) takes a 16-channel Wan VAE latent as input and produces an upscaled 16-channel latent, preserving the temporal dimension:

$$U_\psi^{\text{CLL}} : \mathbb{R}^{B \times 16 \times F \times h \times w} \rightarrow \mathbb{R}^{B \times 16 \times F \times H \times W}.$$

The architecture adopts an encoder-resampler-decoder structure inspired by LTX-2. A 3D convolutional input layer projects the latents into a 256-dimensional hidden space, followed by 4 spatiotemporal residual blocks to extract rich spatiotemporal features. A spatial resampler then scales the resolution, after which 4 additional 3D residual blocks and a final 3D convolutional layer reconstruct the 16-channel output. This spatiotemporal processing enables the model to exploit information from adjacent latent frames while keeping the total frame count invariant.

We implement two distinct spatial resampling configurations. For the  $480 \times 832 \rightarrow 720 \times 1248$  transition, the latent grid expands from  $60 \times 104$  to  $90 \times 156$ . The network first inflates the channel dimension by a factor of 9, applies a  $3 \times$  spatial PixelShuffle, and finally performs a stride-2 downsampling using a fixed binomial blur kernel to achieve a rational  $3/2 \times$  upscaling:

$$60 \times 104 \xrightarrow{\times 3} 180 \times 312 \xrightarrow{/2} 90 \times 156.$$

For the  $368 \times 640 \rightarrow 720 \times 1248$  transition, the latent grid expands from  $46 \times 80$  to  $90 \times 156$ . Since the target size is not an exact integer multiple, the network executes a  $2 \times$  PixelShuffle to reach  $92 \times 160$ , followed by a center crop to isolate the exact  $90 \times 156$  target grid.

The training objective for CLL is defined as:

$$\mathcal{L}_{\text{CLL}} = \mathcal{L}_{\text{rec}} + \lambda_{\text{content}} \mathcal{L}_{\text{content}} + \lambda_{\text{temp}} \mathcal{L}_{\text{temp}},$$

where

$$\mathcal{L}_{\text{rec}} = \mathbb{E} \left[ \sqrt{(\hat{z}_0^H - z_0^H)^2 + \epsilon^2} \right].$$

$$\mathcal{L}_{\text{content}} = \|D(\hat{z}_0^H) - z_0^L\|_1, \quad (1)$$

$$\mathcal{L}_{\text{temp}} = \|\Delta_F \hat{z}_0^H - \Delta_F z_0^H\|_1. \quad (2)$$

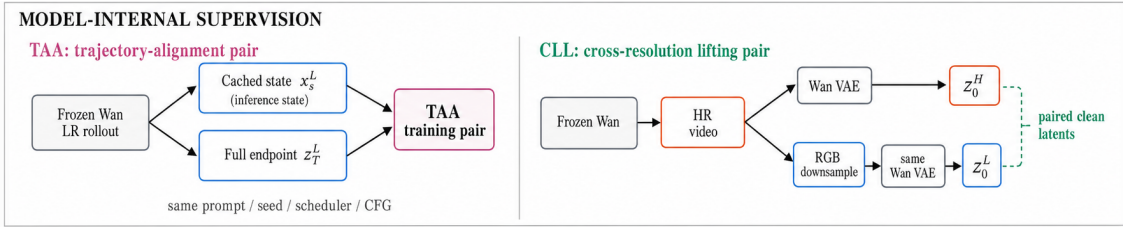


Figure 2: Model-internal supervision for TAA and CLL. TAA pairs the cached handoff state with the complete low-resolution trajectory endpoint under identical prompt, seed, scheduler, and CFG settings. CLL pairs clean latents encoded from a frozen-Wan-generated high-resolution video and its RGB-downsampled counterpart. Both training-pair streams are generated by the same frozen Wan model.

Here,  $D$  denotes a downscaling operator that maps the predicted high-resolution latent back to the low-resolution grid, and  $\Delta_F$  represents the first-order temporal difference between adjacent latent frames. We set  $\lambda_{\text{content}} = 0.2$ ,  $\lambda_{\text{temp}} = 0.1$ , and  $\epsilon = 10^{-3}$  in our experiments. These three loss terms jointly constrain point-wise reconstruction, low-frequency content preservation, and temporal consistency. Because a low-resolution latent cannot uniquely determine high-resolution details, CLL focuses on recovering a stable cross-scale representation, leaving the remaining high-frequency details to be synthesized and corrected by the subsequent HTR and high-resolution diffusion priors.

### 3.4 Handoff-Local Trajectory Alignment

TAA is implemented via LoRA (Hu et al. 2022) and is activated exclusively at the designated handoff step. The base Wan model remains completely frozen; low-rank updates of rank  $r$  are injected solely into the  $q, k, v, o$  projections of the attention modules and the  $fn.0, fn.2$  linear layers of the Feed-Forward Networks (FFNs):

$$W' = W + \frac{\alpha}{r} BA.$$

Under the flow matching parameterization, the TAA-aligned clean latent prediction at step  $s$  is expressed as:

$$\tilde{z}_s^L = x_s^L - \sigma_s v_{\phi + \Delta\theta_s}(x_s^L, s, c).$$

The alignment target is the low-resolution trajectory endpoint  $z_T^L$  obtained by fully running the frozen model up to step  $T$  under identical conditions (text prompt, random seed, initial noise, and scheduler). The trajectory alignment loss is formulated as:

$$\mathcal{L}_{\text{align}} = \|\tilde{z}_s^L - z_T^L\|_1 + 0.1 \|\tilde{z}_s^L - z_T^L\|_2^2.$$

For earlier handoffs (such as step 40), we introduce an additional temporal residual alignment loss with a weight of 0.05 to mitigate the propagation of frame-to-frame errors into the CLL module. While our framework also supports direct supervision via the velocity target  $v^* = \frac{x_s^L - z_T^L}{\sigma_s}$ , all experiments in this paper report results optimized using the clean latent alignment objective above.

By compressing the residual error of the low-resolution tail trajectory at the handoff step, TAA effectively projects the input of CLL back onto the clean latent training domain. While the system-level acceleration of TALH stems from

the low-resolution prefix computations, TAA ensures trajectory consistency at fixed handoff points. Earlier handoffs, which inherently suffer from larger residual trajectory gaps, consequently yield a broader optimization space for TAA calibration.

### 3.5 Cached-Prefix State Consistency

Assuming TAA is activated strictly at the handoff step  $s$ , its LoRA parameters satisfy  $\Delta\theta_k = 0$  for all steps  $k < s$ . Given a deterministic sampler alongside fixed text embeddings, random seeds, schedulers, and Classifier-Free Guidance (CFG) scales, the adapted generative process shares the identical transition functions as the base process prior to step  $s$ . Thus, by induction over the sampling steps, we establish:

$$x_k^{\text{adapted}} = x_k^{\text{base}}, \quad \forall k \leq s.$$

This property guarantees that the pre-cached base model prefix state  $x_s^L$  is mathematically identical to the state actually accessed by TAA during inference. For localized adaptations restricted to a single handoff step, caching the prefix state allows us to bypass redundant prefix computations entirely during inference without introducing any training-inference distribution drift, maintaining an  $O(1)$  state retrieval overhead.

This consistency property relies on four strict conditions: TAA must be engaged exclusively at the handoff step; training and inference must share identical schedulers, CFG scales, and conditional inputs; the prefix sampling path must be entirely deterministic; and no unaligned stochastic operations can occur within the prefix trajectory. If TAA were extended across multiple steps, its parameters would alter subsequent states, requiring advanced training mechanisms such as student model rollouts, synchronized teacher predictions, or Exponential Moving Average (EMA) teacher models.

### 3.6 High-Resolution Trajectory Re-entry and Two Instances

Once TAA produces the aligned low-resolution clean latent, CLL outputs the upsampled high-resolution clean latent:

$$\hat{z}_s^H = U_{\psi}^{\text{CLL}}(\tilde{z}_s^L).$$

The High-Resolution Trajectory Re-entry (HTR) phase then introduces target-resolution noise according to the schedule coefficient  $\sigma_{s+1}$  of the next timestep:

$$x_{s+1}^H = (1 - \sigma_{s+1})\hat{z}_s^H + \sigma_{s+1}\epsilon^H.$$

The remaining denoising calculations from step  $s + 1$  to  $T$  are subsequently completed by the frozen Wan denoiser operating on the high-resolution latent space. Unlike an *Interpolated-Flow Re-entry* strategy that directly interpolates low-resolution velocity flows into the high-resolution space, TALH utilizes *Target-Resolution Noise Re-entry*. This design ensures that newly introduced spatial frequencies are synthesized natively by the high-resolution generative priors of the base model.

Our **50-step pipeline** uses Wan2.1 T2V-1.3B, configured with 50 inference steps, a sample shift of 8, and a CFG scale of 6. Extensive handoff-step scanning identified two representative operational points. The quality-centric **TALH-Q** (TALH@40) reserves 10 high-resolution steps, granting a larger budget for texture generation while managing a wider residual trajectory gap that requires substantial TAA alignment. Conversely, the efficiency-centric **TALH-E** (TALH@45) reserves only 5 high-resolution steps to minimize latency; its handoff state sits closer to a fully converged clean latent, resulting in more stable CLL inputs and narrower TAA corrections. Together, these two instances define the quality-efficiency Pareto frontier of TALH.

Our **4-step pipeline** leverages Wan2.1 T2V-14B StepDistill-CfgDistill, operating on four nominal timesteps [1000, 750, 500, 250] with a sample shift of 5. The distilled instance **TALH-D4** (TALH@3-of-4) runs low-resolution sampling for the first two steps. At step 3, TAA predicts the aligned low-resolution clean latent, which is scaled by CLL and injected back into the high-resolution trajectory via HTR at step 4. A single final high-resolution denoising step completes the generation.

## 4 Experiments

### 4.1 Experimental Setup

We evaluate TALH at both system and module levels, covering end-to-end quality-latency trade-offs, the individual and interactive effects of CLL and TAA, and compatibility with a 4-step distilled sampler.

For the 50-step pipeline, we generate approximately 1,000 reference videos using Wan2.1 T2V-1.3B. Each video contains 81 frames at  $720 \times 1248$  resolution and 16 fps. Bicubic RGB downsampling produces corresponding  $480 \times 832$  and  $368 \times 640$  videos, which are re-encoded by the frozen Wan VAE to construct aligned cross-resolution latent pairs. TAA training pairs consist of the low-resolution state immediately preceding a designated handoff and the endpoint of the corresponding complete low-resolution trajectory.

The 4-step experiments use approximately 5,000 videos generated by Wan2.1 T2V-14B StepDistill-CfgDistill. CLL is trained on  $368 \times 640 \rightarrow 720 \times 1248$  clean latent pairs, while TAA pairs the state preceding step 3 with the complete step-4 low-resolution endpoint. Training, validation, and test partitions are separated by both text prompts and random seeds.

CLL is optimized with AdamW for up to 50k steps using a learning rate of  $1 \times 10^{-4}$ , *bf16* precision, and an EMA

decay of 0.9999. TAA is trained for up to 10k steps with AdamW, a learning rate of  $5 \times 10^{-5}$ , and LoRA rank/scale 32/32. Training uses four NVIDIA H100 GPUs and requires approximately 8 hours for CLL and 33 hours for TAA. All efficiency measurements are averaged over five independent cold-start runs. Since all supervision is generated by the frozen target model, TALH requires neither external paired videos nor an additional upscaling teacher.

### 4.2 Baselines and Evaluation Protocol

System-level comparisons include **Native-HR Sampling**, which executes all denoising steps at 720p; **Trilinear Hand-off**, which performs mixed-resolution inference using fixed trilinear latent interpolation; and **Endpoint Re-entry**, which completes the low-resolution trajectory, applies CLL, and performs one low-intensity high-resolution refinement step. We additionally report **Native-LR Sampling** as a low-resolution reference.

Module interactions are evaluated through a  $2 \times 2$  factorial design that crosses trajectory states {Unaligned, TAA-aligned} with resolution operators {Trilinear, CLL}. The four resulting configurations are Trilinear Handoff, TAA+Trilinear, CLL-only, and the complete TALH framework. Oracle-Aligned CLL and Joint Trajectory-Scale Lifting (JTSL) are used as diagnostic references; extended results are provided in the Supplementary Material.

For trajectory alignment, we report endpoint latent  $L_1$ , decoded PSNR and LPIPS, temporal latent differences, and sample-wise win rates. CLL is evaluated using latent  $L_1$ , PSNR, SSIM, LPIPS, and temporal  $L_1$ . End-to-end quality is measured with VBench (Huang et al. 2024) and human evaluation. We define **VBench-5** as the unweighted mean of subject consistency, background consistency, motion smoothness, aesthetic quality, and imaging quality.

All paired configurations share identical prompts, random seeds, initial noise, schedulers, and guidance parameters. Prompt-level score differences are analyzed using 10,000 paired bootstrap iterations for 95% confidence intervals and two-sided exact sign tests. Human evaluation uses randomized, double-blind comparisons over 10 matched prompts, with three independent judges per prompt and prompt-level majority voting. Inter-rater agreement is reported in the Supplementary Material.

Efficiency measurements include end-to-end cold-start latency, peak GPU memory, low- and high-resolution denoising evaluations, and speedup over Native-HR Sampling. All TAA, CLL, HTR, VAE, and dynamic model-loading overheads are included.

## 5 Results and Analysis

### 5.1 System-Level Quality-Efficiency Trade-off

Table 1 compares the standard 50-step Native-HR pipeline with the two principal TALH operating points. TALH-Q moves the first 40 evaluations to the low-resolution grid and retains 10 high-resolution suffix evaluations, reducing latency from 253.10 to 138.36 seconds (1.83 $\times$ ). TALH-E

retains only five high-resolution evaluations and further reduces latency to 114.26 seconds ( $2.22\times$ ). Their VBench-5 scores are 0.80983 and 0.80792, compared with 0.82836 for Native-HR Sampling.

The Endpoint Re-entry baseline reaches 86.45 seconds ( $2.93\times$ ) but decreases VBench-5 to 0.80093, with inferior scores on 9/10 prompts relative to Native-HR ( $p = 0.02148$ ). TALH therefore retains a longer high-resolution suffix when additional structural and texture refinement is beneficial, while later handoffs prioritize inference speed. Peak memory remains approximately 26 GiB across Native-HR, TALH-Q, TALH-E, and Endpoint Re-entry, indicating that the primary system benefit of TALH is reduced runtime rather than memory consumption.

## 5.2 Component Analysis

**Clean latent lifting.** Table 2 evaluates CLL against trilinear interpolation over 50 paired clean latent samples. For  $480p \rightarrow 720p$ , CLL reduces latent  $L_1$  from 0.1929 to 0.0991, corresponding to a 48.6% reduction. It also improves decoded PSNR by 5.71 dB, reduces LPIPS from 0.2358 to 0.0629, and lowers temporal  $L_1$  by 30.8%. CLL wins on all 50 samples for PSNR, LPIPS, and temporal consistency, and on 49/50 samples for SSIM.

The more aggressive  $368p \rightarrow 720p$  transition shows the same trend: latent  $L_1$ , LPIPS, and temporal  $L_1$  are reduced by 33.4%, 51.1%, and 15.5%, respectively. These consistent latent, perceptual, and temporal improvements indicate that CLL learns a substantially more faithful cross-resolution mapping than fixed interpolation.

**Trajectory alignment.** At an alignment intensity of 0.75, TAA reduces the residual low-resolution endpoint  $L_1$  error by 25.82% at step 40 and 21.03% at step 45. Both settings improve all 10 evaluation samples, with paired confidence intervals excluding zero. TAA also increases decoded PSNR by 2.55 and 2.23 dB and reduces latent temporal difference errors by 23.44% and 21.24%, respectively. On the 4-step distilled model, TAA provides a smaller but consistent 5.03% endpoint-error reduction.

Alignment-intensity validation reveals a non-monotonic dependence on correction strength. An intermediate intensity of 0.75 provides the lowest endpoint error and the strongest downstream quality, whereas weaker or full-strength corrections are less effective. This behavior supports a calibrated rather than unconstrained trajectory correction.

**Factorial effects.** The  $2 \times 2$  analysis in Table 4 shows that CLL provides the principal broad quality gain, improving VBench-5 by approximately 0.031–0.041 across the three evaluated pipelines. TAA serves a more targeted role. With CLL fixed, it improves the imaging-quality dimension by +0.00599, +0.00254, and +0.00479 at step 40, step 45, and on the 4-step sampler, respectively. These detail-oriented improvements are distributed across only one component of the five-dimensional aggregate, and therefore appear as smaller changes in the averaged VBench-5 score.

The results support a clear division of labor. CLL restores the cross-resolution latent representation and supplies the dominant improvements in motion smoothness, aesthetics,

and overall imaging quality. TAA further calibrates the hand-off state toward the endpoint trajectory, enriching localized textures and fine structures that may be underrepresented by aggregate automatic metrics.

## 5.3 Perceptual Effects and Distilled-Sampler Compatibility

Human evaluation confirms that the detail recovery introduced by TAA is perceptually visible. With the same CLL operator, TAA receives fine-detail win/loss/tie counts of 8/2/0 at step 40 and 9/0/1 at step 45. Overall-quality judgments are also positive, reaching 6/2/2 and 6/0/4, respectively. These preferences support the dimension-level VBench observation that TAA selectively improves imaging quality and localized structure.

The additional fine-scale content introduces a spatial-temporal trade-off: temporal stability favors the unaligned CLL state in both settings. This suggests that TAA shifts the operating point toward sharper local rendering, while CLL and the remaining high-resolution suffix provide the broader spatial and temporal reconstruction.

CLL provides the broader perceptual improvement: at step 45, both CLL-only and TALH-E obtain 10/10 prompt-level wins over trilinear handoff for fine details and overall quality, while also improving temporal stability. TAA adds a narrower refinement of local detail on top of this cross-resolution reconstruction.

TALH is also compatible with the 4-step distilled Wan sampler. On the 368p trajectory, Trilinear Handoff achieves a VBench-5 score of 0.81796, whereas TALH-D4 reaches 0.85680. The associated cold-start latency increases only from 167.09 to 170.29 seconds. As in the 50-step setting, CLL supplies the primary cross-resolution quality improvement, while TAA performs localized endpoint calibration and improves the imaging-quality component.

Qualitative comparisons in Figure 1(a) further illustrate CLL’s recovery of high-frequency textures and TAA’s localized adjustments. Extended spatial crops and temporal sequences are provided in Figure S1. The content-matched Native-HR (estimated) reference is produced from a complete 368p trajectory and is used only for frame-level structural comparison; true 720p Native-HR Sampling is used for the quantitative evaluation in Table 1.

## 6 Discussion and Limitations

TALH assigns trajectory correction, cross-resolution lifting, and high-resolution refinement to TAA, CLL, and HTR. This decomposition keeps CLL in a stable clean-latent domain while allowing the frozen generator to recover high-frequency details after re-entry. Endpoint metrics, imaging-quality gains, and human preferences indicate that TAA improves local detail and overall perceptual quality, although its effect depends on the CLL input distribution and high-resolution suffix.

TAA and CLL use model-endogenous self-distillation: the frozen Wan model supplies high-resolution references for CLL and complete low-resolution trajectories for TAA,



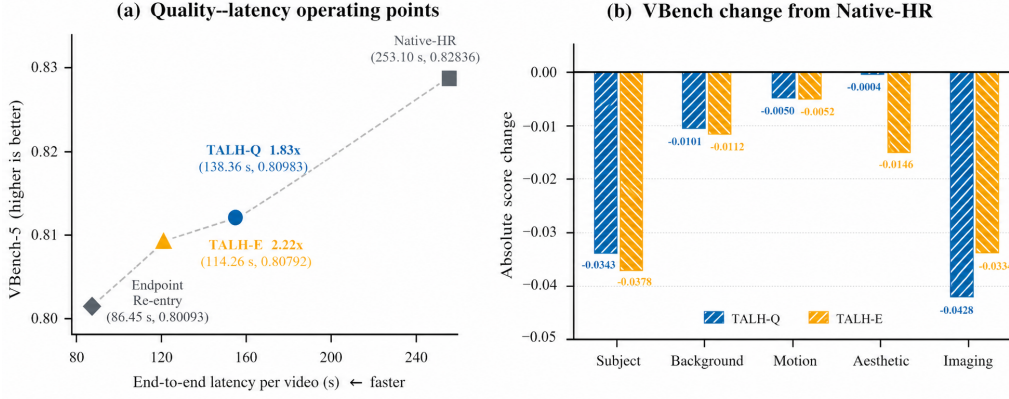


Figure 3: Quality-efficiency trade-offs relative to Native-HR Sampling. (a) TALH-Q and TALH-E provide quality- and efficiency-oriented operating points, while Endpoint Re-entry achieves higher speed at a larger quality cost. (b) Absolute per-dimension VBench changes relative to Native-HR.

| Method             | LR/HR evals. | Time (s) ↓           | Latency red.  | Speedup ↑    | VBench-5 ↑     | Retention ↑ |
|--------------------|--------------|----------------------|---------------|--------------|----------------|-------------|
| Native-HR Sampling | 0/50         | 253.10 ± 1.53        | —             | 1.00×        | <b>0.82836</b> | 100.00%     |
| TALH-Q (Step 40)   | 40/10        | 138.36 ± 0.50        | 45.33%        | 1.83×        | 0.80983        | 97.76%      |
| TALH-E (Step 45)   | 45/5         | <b>114.26 ± 1.52</b> | <b>54.85%</b> | <b>2.22×</b> | 0.80792        | 97.53%      |

Table 1: System-level comparison against Native-HR Sampling. Timings average five cold-start runs, and quality is evaluated on ten matched prompts with all processing overheads included.

| Input → Output | Operator  | Latent $L_1$ ↓ | PSNR ↑       | SSIM ↑        | LPIPS ↓       | Temp. $L_1$ ↓  |
|----------------|-----------|----------------|--------------|---------------|---------------|----------------|
| 480p → 720p    | Trilinear | 0.1929         | 24.29        | 0.7159        | 0.2358        | 0.02121        |
| 480p → 720p    | CLL       | <b>0.0991</b>  | <b>30.00</b> | <b>0.8696</b> | <b>0.0629</b> | <b>0.01468</b> |
| 368p → 720p    | Trilinear | 0.2444         | 23.42        | 0.6740        | 0.3352        | 0.02388        |
| 368p → 720p    | CLL       | <b>0.1628</b>  | <b>24.10</b> | <b>0.7592</b> | <b>0.1640</b> | <b>0.02017</b> |

Table 2: CLL versus trilinear interpolation on paired clean latents ( $n = 50$ ).

| Trajectory / handoff  | Unaligned $L_1$ ↓ | TAA $L_1$ ↓    | Reduction | Win/Loss |
|-----------------------|-------------------|----------------|-----------|----------|
| Wan50, Steps 40 → 50  | 0.03215           | <b>0.02385</b> | 25.82%    | 10/0     |
| Wan50, Steps 45 → 50  | 0.02363           | <b>0.01866</b> | 21.03%    | 10/0     |
| Distill4, Steps 3 → 4 | 0.04286           | <b>0.04070</b> | 5.03%     | 10/0     |

Table 3: Paired trajectory-alignment results. All three comparisons have two-sided exact sign-test  $p = 0.00195$ ; full confidence intervals are reported in the Supplementary Material.

| Sampler / handoff        | Trilinear | TAA+Tri. | CLL-only       | TALH           | CLL gain | TAA gain | Total gain |
|--------------------------|-----------|----------|----------------|----------------|----------|----------|------------|
| Wan50 / Step 40 (TALH-Q) | 0.77812   | 0.77901  | 0.80896        | <b>0.80983</b> | +0.03085 | +0.00087 | +0.03171   |
| Wan50 / Step 45 (TALH-E) | 0.76776   | 0.76734  | <b>0.80842</b> | 0.80792        | +0.04066 | -0.00050 | +0.04016   |
| Distill4 / Step 3-of-4   | 0.81796   | 0.81909  | 0.85670        | <b>0.85680</b> | +0.03874 | +0.00010 | +0.03884   |

Table 4: End-to-end factorial results on VBench-5 ( $n = 10$ ). “TAA gain” denotes the marginal effect of enabling TAA with CLL fixed.

| Comparison (former vs. latter)        | Details (W/L/T) | Overall (W/L/T) | Temporal (W/L/T) |
|---------------------------------------|-----------------|-----------------|------------------|
| TAA vs. Unaligned, both CLL (Step 40) | 8/2/0           | 6/2/2           | 1/9/0            |
| TAA vs. Unaligned, both CLL (Step 45) | <b>9/0/1</b>    | <b>6/0/4</b>    | 0/8/2            |
| CLL-only vs. Trilinear (Step 45)      | <b>10/0/0</b>   | <b>10/0/0</b>   | <b>8/0/2</b>     |
| TALH-E vs. Trilinear (Step 45)        | <b>10/0/0</b>   | <b>10/0/0</b>   | <b>9/0/1</b>     |

Table 5: Prompt-level majority votes over ten prompts with three raters per prompt. Bold entries have two-sided exact sign-test  $p < 0.05$ .

avoiding external paired data, standalone upscalers, and additional teachers.

Several limitations remain. CLL is tied to the target VAE, generator distribution, and scaling configuration, so cross-

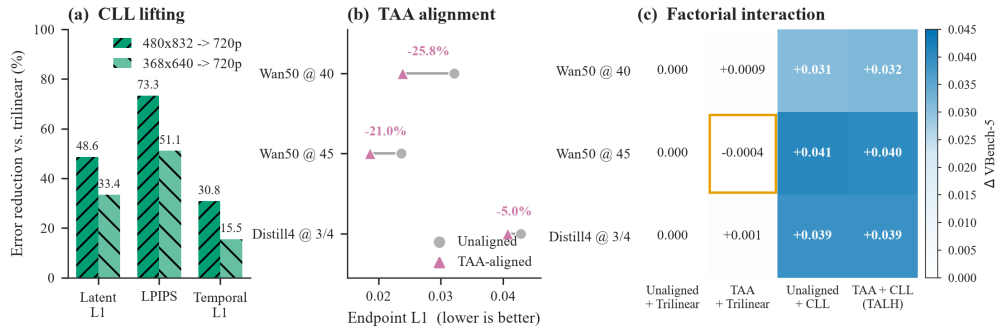


Figure 4: Module effects in TALH. CLL consistently improves latent, perceptual, and temporal reconstruction over trilinear interpolation; TAA narrows the low-resolution trajectory-endpoint gap; and the factorial analysis separates their complementary roles in final video generation.

model transfer is unverified. A distribution gap persists between its training inputs and native low-resolution trajectory states. Fixed handoff steps also require step-specific TAA parameters, motivating timestep-conditioned adapters for adaptive handoffs. Finally, TALH addresses in-trajectory spatial scaling rather than real-world degradations such as compression, blur, or sensor noise. It inherits the base generator’s content risks and safety requirements.

## 7 Conclusion

We presented TALH, a lightweight framework combining a low-resolution prefix with a short high-resolution suffix. TAA aligns the residual trajectory, CLL lifts clean latents across resolutions, and HTR re-enters the frozen high-resolution trajectory, with supervision generated entirely by the base model.

TALH-Q and TALH-E achieve  $1.83\times$  and  $2.22\times$  speedups while retaining 97.76% and 97.53% of Native-HR VBench-5 performance. CLL consistently outperforms trilinear interpolation, while TAA improves residual alignment and localized visual detail across standard and distilled samplers. TALH thus provides a lightweight, controllable quality-efficiency trade-off for high-resolution video generation.

## References

Blattmann, A.; Rombach, R.; Ling, H.; Dockhorn, T.; Kim, S. W.; Fidler, S.; and Kreis, K. 2023. Align Your Latents: High-Resolution Video Synthesis with Latent Diffusion Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.

Gu, Y.; Fang, G.; Jiang, Y.; Mao, W.; Han, S.; Cai, H.; and Shou, M. Z. 2026. AnyFlow: Any-Step Video Diffusion Model with On-Policy Flow Map Distillation. arXiv:2605.13724.

HaCohen, Y.; Brazowski, B.; Chiprut, N.; Bitterman, Y.; Kvochko, A.; Berkowitz, A.; Shalem, D.; Lifschitz, D.; Moshe, D.; Porat, E.; Richardson, E.; Shiran, G.; Chachy, I.; Chetboun, J.; Finkelson, M.; Kupchick, M.; Zabari, N.; Guetta, N.; Kotler, N.; Bibi, O.; Gordon, O.; Panet, P.; Benita, R.; Armon, S.; Kulikov, V.; Inger, Y.; Shiftan, Y.; Melumian, Z.; and Farbman, Z. 2026. LTX-2: Efficient Joint Audio-Visual Foundation Model. arXiv:2601.03233.

Hu, E. J.; Shen, Y.; Wallis, P.; Allen-Zhu, Z.; Li, Y.; Wang, S.; Wang, L.; and Chen, W. 2022. LoRA: Low-Rank Adaptation of Large Language Models. In *International Conference on Learning Representations*.

Huang, X.; Li, Z.; He, G.; Zhou, M.; and Shechtman, E. 2025. Self Forcing: Bridging the Train-Test Gap in Autoregressive Video Diffusion. arXiv:2506.08009.

Huang, Z.; He, Y.; Yu, J.; Zhang, F.; Si, C.; Jiang, Y.; Zhang, Y.; Wu, T.; Jin, Q.; Chanpaisit, N.; Wang, Y.; Chen, X.; Wang, L.; Lin, D.; Qiao, Y.; and Liu, Z. 2024. VBench: Comprehensive Benchmark Suite for Video Generative Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.

Jeong, W.; Lee, K.; Seo, H.; and Chun, S. Y. 2025. Training-Free Mixed-Resolution Latent Upsampling for Spatially Accelerated Diffusion Transformers. arXiv:2507.08422.

Luo, S.; Tan, Y.; Huang, L.; Li, J.; and Zhao, H. 2023. Latent Consistency Models: Synthesizing High-Resolution Images with Few-Step Inference. arXiv:2310.04378.

ModelTC. 2026. LightX2V: Variable Resolution Inference and Step Distillation Documentation. <https://github.com/ModelTC/LightX2V>. Accessed: 2026-07-15.

Rombach, R.; Blattmann, A.; Lorenz, D.; Esser, P.; and Ommer, B. 2022. High-Resolution Image Synthesis with Latent Diffusion Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.

Wan, T.; Wang, A.; Ai, B.; Wen, B.; Mao, C.; Xie, C.-W.; Chen, D.; Yu, F.; Zhao, H.; Yang, J.; Zeng, J.; Wang, J.; Zhang, J.; Zhou, J.; Wang, J.; Chen, J.; Zhu, K.; Zhao, K.; Yan, K.; Huang, L.; Feng, M.; Zhang, N.; Li, P.; Wu, P.; Chu, R.; Feng, R.; Zhang, S.; Sun, S.; Fang, T.; Wang, T.; Gui, T.; Weng, T.; Shen, T.; Lin, W.; Wang, W.; Wang, W.; Zhou, W.; Wang, W.; Shen, W.; Yu, W.; Shi, X.; Huang, X.; Xu, X.; Kou, Y.; Lv, Y.; Li, Y.; Liu, Y.; Wang, Y.; Zhang, Y.; Huang, Y.; Li, Y.; Wu, Y.; Liu, Y.; Pan, Y.; Zheng, Y.; Hong, Y.; Shi, Y.; Feng, Y.; Jiang, Z.; Han, Z.; Wu, Z.-F.; and Liu, Z. 2025. Wan: Open and Advanced Large-Scale Video Generative Models. arXiv:2503.20314.

Xu, Y.; Park, T.; Zhang, R.; Zhou, Y.; Shechtman, E.; Liu, F.; Huang, J.-B.; and Liu, D. 2024. VideoGigaGAN: Towards Detail-Rich Video Super-Resolution. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.



Yin, T.; Gharbi, M.; Park, T.; Zhang, R.; Shechtman, E.; Durand, F.; and Freeman, W. T. 2024. Improved Distribution Matching Distillation for Fast Image Synthesis. arXiv:2405.14867.

Zhai, Y.; Lin, K.; Yang, Z.; Li, L.; Wang, J.; Lin, C.-C.; Doermann, D.; Yuan, J.; and Wang, L. 2024. Motion Consistency Model: Accelerating Video Diffusion with Disentangled Motion-Appearance Distillation. arXiv:2406.06890.

Zhao, C.; Chen, J.; Li, H.; Kang, Z.; Lu, S.; Wei, X.; Zhang, K.; Yang, J.; and Tai, Y. 2026. LUVE: Latent-Cascaded Ultra-High-Resolution Video Generation with Dual Frequency Experts. arXiv:2602.11564.

Zheng, X.; Liu, X.; Ding, Y.; Feng, W.; Lin, J.; Guo, J.; and Qin, H. 2026. Multi-Resolution Flow Matching: Training-Free Diffusion Acceleration via Staged Sampling. arXiv:2607.01642.

Zhou, S.; Yang, P.; Wang, J.; Luo, Y.; and Loy, C. C. 2024. Upscale-A-Video: Temporal-Consistent Diffusion Model for Real-World Video Super-Resolution. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.